U.S. ELECTRICAL GRID

Intelligence & Operations Dashboard

Data Source Report, Key Findings & Business Value

Data Year: 2024 **| Source Agency:** U.S. Energy Information Administration (EIA)
Database: SQL Server Express — EnergyWorkforce_DB
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1. Data Sources & Coverage

This dashboard is built exclusively on publicly available data published by the U.S. Energy Information Administration (EIA). All data reflects the 2024 reporting year covering the 50 U.S. states plus the District of Columbia.

1.1 Primary Data Sources

Source File	Description & Coverage
EIA Form 861 — Sales_2024.csv	Annual Electric Power Industry Report. Covers utility-level revenue, electricity sales (MWh), and customer counts by sector: Residential, Commercial, Industrial, and Transportation.
EIA Form 861 — Reliability_2024.csv	Reliability metrics reported by electric utilities. Includes SAIDI, SAIFI, CAIDI, number of customers, outage events, and voltage data under both IEEE and Other standard methodologies.
Operational_Data_2024.csv	Utility-level power generation, wholesale purchases, energy losses, transmission data, peak demand (summer/winter), and revenue breakdown by source.
EIA Form 930 — EIA930_BALANCE / INTERCHANGE files	Hourly grid balance and interchange data by Balancing Authority (2025–2026). Collected for future trend work; not used in the 2024 dashboard KPIs.
QCEW_2211_2024_All_States.csv	Quarterly Census of Employment and Wages. Collected as supplemental workforce context; not shown in the dashboard.
nat4d_M2023 / natsector_M2023	National employment and sector data for energy workforce benchmarking. Collected as supplemental context; not shown in the dashboard.

1.2 Data Year & Reporting Period

All dashboard metrics reflect calendar year 2024 data from EIA Form 861 and the EIA operational file. The EIA Form 930 files (2025–2026) and the workforce files were collected for future analysis and are not part of the dashboard KPIs.

1.3 Data Pipeline & Model Architecture

Raw CSV and Excel files were cleaned, transformed, and loaded into a SQL Server Express database (EnergyWorkforce_DB). The Power BI semantic model imports the operational and sales fact tables from this database and the reliability fact table directly from the EIA-861 reliability file (see Section 5), arranged as a star schema:

Layer	Tables & Views
Fact Tables	fact_operational, fact_reliability, fact_sales
Dimension Tables	dim_nerc_region, dim_ownership_type, dim_state
SQL Views	vw_reliability_by_region, vw_revenue_efficiency, vw_state_energy_summary, vw_utility_sales_by_sector
DAX Measures	35 custom DAX measures covering generation, reliability, revenue, risk scoring, and efficiency KPIs
Records Covered	About 5,150 utility-level records (1,370 operational, 2,814 sales, 971 reliability) across 50 states and DC

2. Dashboard Pages & Key Findings

The dashboard is organized into four analytical domains, each targeting a distinct operational dimension of the U.S. electrical grid.

PAGE 1 — OPERATIONS

What It Shows

Grid-level net generation, wholesale purchases, total supply, energy losses, summer peak demand, generation by state and NERC region, and the generation mix by ownership type.

Key Findings

- U.S. utilities reported 2.09 billion MWh of net generation in 2024. Florida leads all states at 251 million MWh, followed by North Carolina (161M), Oregon (121M), Georgia (118M) and Texas (108M).
- Wholesale purchases total 2.95 billion MWh, more than the utilities generate themselves: 58% of all supply is bought rather than generated, which is why purchase dependency drives the Supply Vulnerability Score.
- Total supply (generation plus purchases and exchanges) reached 5.06 billion MWh. By region, SERC generates the most at 674 million MWh, followed by WECC (450M) and FRCC (251M).
- Energy losses totaled 168 million MWh, a 3.3% loss rate on total supply, flagged NORMAL by the Loss Rate Risk model (thresholds: 5% elevated, 8% high risk).
- Investor-owned utilities produce 67.7% of net generation, followed by Cooperatives (10.1%), Political Subdivisions (6.6%), Municipals (5.6%), State utilities (3.5%) and Federal utilities (3.3%).
- Summer peak demand totals 962,171 MW against a winter peak of 848,643 MW, giving a Seasonal Load Imbalance score of 12.5 — a moderate, summer-driven gap.

PAGE 2 — REVENUE & SALES

What It Shows

Total utility revenue, revenue per MWh, revenue per customer, customer count, sector revenue mix (residential, commercial, industrial, transportation), revenue by ownership type and state, and a state × sector revenue matrix.

Key Findings

- Total U.S. retail electricity revenue for 2024 reached $514.4 billion. Investor-owned utilities earned $298.6 billion (58%), followed by Retail Power Marketers ($71.2B) and Cooperatives ($55.2B).
- Revenue per MWh averages $112.59 nationally, ranging from $64 in Ohio and $76 in Illinois to $198 in California and $380 in Hawaii.
- Revenue per customer averages $2,838 across 181.2 million customer accounts. Hawaii and Alaska stand out on price because of geographic isolation and fuel-import costs.
- Residential customers generate the largest share of revenue at 47.5%, followed by Commercial (36.0%), Industrial (16.3%) and Transportation (0.2%) — a consumer-driven revenue base.
- California ($66.4B) and Texas ($49.5B) lead state revenue, followed by Florida ($32.0B), New York ($27.6B) and Pennsylvania ($17.8B).
- The Industry Risk flag reads HEALTHY nationally: industrial revenue is 16.3% of the total, well below the 25% moderate and 35% high-risk thresholds.

PAGE 3 — RELIABILITY

What It Shows

Utility and regional outage performance measured by SAIDI (minutes without power per customer per year), SAIFI (interruptions per customer) and CAIDI (minutes per interruption), all weighted by customers served. Separates storm-driven outages (major event days) from controllable outages and ranks the worst-performing utilities.

Key Findings

- The average customer lost power for 662.6 minutes (about 11 hours) in 2024 including major events. Excluding major event days this drops to 131.6 minutes, so storms caused about 80% of all outage time in a year marked by Hurricanes Beryl, Helene and Milton.
- Customers averaged 1.53 interruptions in 2024, and each interruption lasted 432.7 minutes on average (CAIDI) — restoration after major storms, not outage frequency, drives the total.
- Grid Stability Score is 73.7 out of 100. The score measures controllable reliability: 100 minus a fifth of the customer-weighted SAIDI excluding major events (500 minutes or more scores 0), so storm damage is reported separately rather than hiding day-to-day performance.
- ERCOT (Texas) and FRCC (Florida) carry the largest storm impact: 1,614 and 1,488 minutes of SAIDI including major events, versus only 153 and 67 minutes of controllable outage. SERC follows at 1,097 minutes.
- The Top 20 Worst Utilities table is led by Altamaha EMC (GA, 17,313 minutes), Jefferson EMC (GA, 10,318) and Haywood EMC (NC, 9,612) — all hit by Hurricane Helene in September 2024.
- Cooperatives have the weakest controllable reliability (188 minutes excluding major events, versus 128 for investor-owned and 90 for municipal utilities), consistent with long rural lines and less capital for hardening.

PAGE 4 — INTELLIGENCE

What It Shows

Grid risk analysis: a Supply Vulnerability Score (60% purchase dependency, 40% loss rate), seasonal load imbalance, loss-rate risk flags, a state risk scatter, and a decomposition tree from region to ownership type, state and utility.

Key Findings

- The national Supply Vulnerability Score is 36.3 out of 100 — moderate, and driven by purchase dependency (58% of supply is bought) rather than losses (3.3%).
- Seasonal Load Imbalance is 12.5: summer peak demand runs about 13% above winter, so capacity planning is summer-driven.
- The national loss rate is 3.3% (NORMAL), but 328 of 1,370 utilities exceed 5%. Georgia Transmission Corp and Vermont Electric Power Co show 100% because they are transmission-only businesses with no retail sales, not because their networks fail.
- By region, NPCC (New England and New York) scores highest on Supply Vulnerability at 54.3, followed by ERCOT (50.6) and RFC (41.6). FRCC is lowest at 11.8 because Florida utilities generate most of their own power.
- The risk scatter places four states in the high-risk corner (loss rate above 6% and purchase dependency above 40%): Rhode Island (19.4% loss, 100% purchased), New Mexico (11.5%, 73%), New Hampshire (7.4%, 100%) and Hawaii (6.2%, 41%).
- The decomposition tree lets users trace each regional score down to ownership type, state and individual utility to find where the dependency sits.

3. Who This Report Is For

This dashboard and its findings are designed to serve five distinct stakeholder groups, each with different information needs and decision-making authority:

Stakeholder	Primary Use Case	Key Pages
Grid Operators & ISOs	Monitor real-time and annual generation balance, regional supply shortfalls, and seasonal load imbalance requiring capacity adjustments.	Operations, Intelligence
Utility Executives	Track revenue performance, customer growth, sector mix, and ownership-type benchmarking to guide investment and rate strategy.	Revenue & Sales, Operations
Energy Regulators (FERC, State PUCs)	Assess reliability compliance (SAIDI/SAIFI/CAIDI targets), identify chronic underperformers, and evaluate storm hardening progress.	Reliability, Intelligence
Infrastructure Investors & Private Equity	Evaluate state-level revenue efficiency, ownership concentration risk, and vulnerability scores before capital deployment decisions.	Revenue & Sales, Intelligence
Policy Analysts & DOE	Identify regional equity gaps in reliability, purchase dependency concentrations, and states most exposed to supply chain disruption.	Intelligence, Reliability

4. Business Value & Practical Benefits

4.1 Operational Benefits

- Replaces manual state-by-state spreadsheet analysis by consolidating about 5,150 utility records into a single interactive dashboard.
- NERC Region and Ownership Type slicers filter every card and chart on all four pages, so operators can isolate any region or utility class across all four analytical domains at once.
- The Supply Vulnerability Score and Loss Rate Risk Flag provide automated early-warning signals that previously required manual calculation and interpretation.

4.2 Financial Benefits

- Revenue per MWh benchmarking enables utilities to identify pricing efficiency gaps relative to peer states and ownership types.
- The Industrial Dependency Flag prevents overexposure risk — utilities above 35% industrial revenue concentration face significant volatility if large industrial customers reduce consumption.
- State x Sector Revenue Matrix enables granular revenue attribution that supports rate case filings and regulatory submissions.

4.3 Reliability & Risk Benefits

- SAIDI/SAIFI/CAIDI tracking against regional peers allows reliability managers to objectively prioritize capital spending on infrastructure hardening.
- Storm vs. Controllable SAIDI separation reveals whether outage problems are weather-driven (requiring physical hardening investment) or operationally driven (requiring process improvement).
- The Top 20 Worst Utilities ranking provides regulators with an evidence-based watchlist for compliance follow-up and targeted performance improvement orders.

4.4 Strategic & Intelligence Benefits

- The Risk Drill-Down decomposition tree enables executives to trace national risk scores down to specific utility level in seconds — a capability that previously required multiple separate analytical tools.
- Purchase Dependency mapping identifies states and regions where grid security is most exposed to wholesale market disruption or counterparty failure.
- Regional vulnerability scoring supports long-term transmission planning, particularly for import-dependent regions such as New England and New York (NPCC) and storm-exposed regions such as Texas, Florida and the Southeast.

5. Data Limitations & Notes

- All data reflects self-reported utility submissions to the EIA. Accuracy depends on utility compliance and reporting quality.
- Reliability metrics (SAIDI/SAIFI/CAIDI) cover the 747 utilities that report under the IEEE standard with a customer count (about 130 million customers). National and regional figures are weighted by customers served.
- Some utility-level loss rates of 100% reflect transmission-only business models (no retail sales) rather than infrastructure failure. They are correctly flagged HIGH RISK by the formula but need context when interpreted.
- NERC region filters use each state's primary region from the dim_state table (for example, DC, MD and PA map to RFC). The Intelligence utility table still shows each utility's own reported region.
- EIA Form 930 balance files (2025–2026) and the workforce files are not used in the 2024 KPIs.
- Data-quality fix: the SQL copy of the reliability table was loaded with commas inside quoted values split into extra columns (for example "Northern Lights, Inc" and "39,641"). That shifted 136 rows and wiped customer counts in all but 3 rows, so earlier reliability KPIs were based on 3 small co-ops. The model now reads the EIA-861 reliability file directly with correct CSV quoting.
- Operational rows with no reported NERC region (220 rows, about 0.26 billion MWh of generation) and the one legacy ECAR row are excluded by a report-level filter, as are 12 Canadian (CN) utility rows, so the dashboard covers the 50 states and DC only.
- Grid Stability Score is a portfolio-defined index, not an EIA or IEEE metric: 100 × (1 − SAIDI excluding major events ÷ 500 minutes), floored at 0.

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